

Activity 5 - Develop the Integrated Schedule and Cost Baseline

Course: Effective Project Management for Small Projects (TGS-2020505545) Topic: 2 | Develop a Project Plan
Learning outcome: LO2 Duration: 55 minutes Team size: 3-4 learners Tool set: Network Diagram, Gantt Schedule and Budget versus Actual

Objective

Sequence, estimate and fund the authorised scope as one realistic plan (A2, K2).

The situation

The launch must precede Lunar New Year trading. The vendor has proposed a date without dependency logic, resource calendars or reserves. Finance requires a time-phased baseline within S\$180,000.

What you will do

Sequence work packages, calculate the critical path and float, estimate duration and cost using explicit methods, set milestones and reconcile schedule, budget and reserve.

What you will produce

A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.

Tool files in this folder

- [`tools/28 Schedule Network Diagram.xlsx`](#)
- [`tools/29 - Gantt Chart - Schedule - Daily Weekly Monthly and Phases.xlsx`](#)
- [`tools/32 Project Budget - Planned vs Actual - Weekly and Monthly.xlsx`](#)

Data files

- [`data/planning-input.csv`](#)

Step-by-step instructions

1. Translate WBS work packages into activities and identify mandatory and discretionary dependencies.
2. Add finish-to-start, start-to-start, leads and lags only where the logic is defensible.
3. Estimate durations using analogous, parametric, bottom-up or three-point methods and record the basis.
4. Perform forward and backward passes; calculate total float and identify the current critical path.
5. Build the Gantt with decision milestones, resource calendars and acceptance gates.
6. Estimate cost bottom-up and cross-check with an analogous or parametric estimate.
7. Time-phase labour, vendor, equipment, training and contingency; keep management reserve separate.
8. Reconcile every scheduled work package to the budget and every milestone to acceptance evidence.

Evidence to submit

Complete [WORKSHEET.md](#) and attach/export the completed tool files named above.

Acceptance criteria

The dependency network calculates correctly, the plan fits the five-month limit, all authorised work is time-phased and funded, and baseline plus contingency stays within S\$180,000.

Debrief

- A Gantt is the presentation of schedule logic, not the logic itself.
- The critical path can change after every update or approved change.
- Baseline and forecast must remain separate so variance can be explained.

The full procedure is also reproduced in the Learner Guide, Activity 5. The trainer deck contains only the briefing and success criteria.